Project Strategy

AI-Powered Smart Waste Segregation System Using Edge Computing and IoT

Course	Project Planning and Management	**Project Duration**	September 2026 - June 2027
Supervisor	Modassir Ishfaq	**Prepared By**	Zeeshan Ali (FA23-BCE-128)
Group Members	Muhammad Sarmad (FA23-BCE-069), Abdur Rafay (FA23-BCE-008), Zeeshan Ali (FA23-BCE-128)	**Programme**	BS Computer Engineering - Final Year Project

Strategy at a Glance

Pointer	Summary
Need	Manual, point-of-disposal waste segregation is almost absent in Pakistan, so recoverable recyclables are lost.
Vision	Affordable, locally deployable AI that makes correct waste sorting the default at the point of disposal.
Mission	Build and evaluate an offline, on-device smart bin that classifies, sorts and reports status by June 2027.
Size	Small academic project: 3 members, 9 months, one prototype bin, four waste categories.
Strategy	Hybrid: predictive at the macro level (fixed FYP milestones), iterative within each technical phase.
Phases	Foundation, Build, Integrate, Evaluate and Deliver.
Budget	Indicative total of about PKR 72,000 including a 10% contingency (Section 12).

1. Introduction

This report presents the Project Strategy for the Final Year Project (FYP) “AI-Powered Smart Waste Segregation System Using Edge Computing and IoT.” It applies the core strategic project-management concepts taught in *Project Planning and Management* - need, concept, vision and mission, objectives, project size, stakeholders, expertise, strategy, phases, deliverables and budget - to the FYP defined in the approved proposal.

A project strategy is the high-level plan that links a project’s purpose to the way it will be delivered. The proposal describes *what* the system does and *how* it works technically; this document describes how the project will be planned, organised, governed and steered to a successful outcome across its nine-month lifecycle.

2. Project Need and Background

Pakistan generates an estimated 49.6 million tons of solid waste every year, yet recycling rates stay low and an estimated USD 1.2 billion of recoverable value is lost annually. Lahore alone produces roughly 2,200 tons of municipal waste per day, and only about 27% of dry recyclable material is recovered, almost entirely through informal channels. *(Figures as cited in the approved FYP proposal.)*

The root cause identified in the proposal is that segregation at the point of disposal is virtually absent and remains fully manual. The project responds with a low-cost smart bin that classifies waste on-device (Raspberry Pi 4 with a lightweight CNN such as MobileNetV2 or a YOLO-nano variant), sorts it mechanically, and reports bin status on a local IoT dashboard. Three values must be protected throughout planning: **affordability**, **offline reliability** and **local deployability**.

3. Project Concept and Long-Term Outlook

Concept. A user drops an item into the inlet; a camera and an inlet sensor detect it; a quantised CNN running on the Raspberry Pi 4 assigns one of four classes (biodegradable, recyclable, non-recyclable, hazardous); a servo and stepper mechanism route the item to the matching compartment; ultrasonic sensors measure fill levels; and a Flask dashboard shows status on the local network. Low-confidence items fall back to an unsorted compartment.

Long-term view. The FYP delivers Stage 1 of a longer path. Later stages are aspirational and are outside the FYP scope; they are shown to explain the direction the design should not block.

Horizon	Focus	Outcome
Short term (FYP, to Jun 2027)	Working prototype with measured accuracy, latency and power	Demonstrated feasibility of offline, on-device sorting
Medium term (about 1-2 years)	Larger local dataset, improved enclosure, pilot in a campus or institution	Field-tested bin with real-world accuracy data
Long term (about 3-5 years)	Multi-bin network, cloud/mobile options, integration with municipal bodies such as LWMC	Scalable, deployable product; possible publication or competition entry

4. Vision and Mission Statement

4.1 Vision

> *"A Pakistan where waste is sorted correctly at the moment of disposal, through affordable, locally deployable AI that any household, institution or community can adopt."*

4.2 Mission

> *"To design, build and evaluate an offline-capable smart bin that classifies waste on-device, sorts it automatically and reports its status over a local dashboard, proving feasibility against measurable benchmarks by June 2027."*

4.3 Companies and Their Visions - What We Learn

Well-run organisations express their direction in one short, outcome-focused sentence. Three widely known examples (paraphrased from their public statements) show the pattern our vision follows:

Company	Direction in brief	Lesson applied to this project
Tesla	Speed up the world's shift to sustainable energy	Tie the product to a larger sustainability goal - here SDGs 11, 12 and 13
Microsoft	Empower every person and organisation to achieve more	Aim at everyday users, not specialists - households and institutions
Google	Organise the world's information and make it useful to all	State one clear, bounded problem - here, source-level waste segregation

5. Project Size and Scale

Sizing the project early keeps scope, effort and budget realistic. The effort figure below is a planning estimate, assuming about 10 hours per member per week alongside coursework over roughly 40 weeks.

Dimension	Estimate	Dimension	Estimate
Duration	9 months (Sep 2026 - Jun 2027)	Team size	3 members
Effort	About 1,200 person-hours	Deliverable	One working prototype bin
Scope breadth	4 waste classes, 1 bin, 1 dashboard	Cost class	Low (under PKR 100,000)
Complexity	Medium: hardware-software integration and embedded ML	Category	Small academic hardware/software project

6. Strategic Objectives (SMART)

The five proposal objectives are restated in SMART form. Numeric targets are proposed benchmarks anchored to the literature cited in the proposal (about 94-95% accuracy and sub-0.1 s latency) and are to be confirmed with the supervisor.

#	Objective and specific outcome	Measurable target	Achievable and relevant	Time-bound
1	**On-device classification.** CPU-only CNN for 4 waste classes on Raspberry Pi 4.	At least 90% accuracy on a held-out test set; per-class precision and recall reported.	MobileNetV2 and public datasets (TrashNet, TACO) make this feasible; core of the system.	Feb 2027
2	**Real-time, offline inference.** Fully functional with zero internet.	Model inference of about 100 ms or less per item; full sort cycle within 5 s; verified by timed bench tests.	TFLite quantisation reduces latency; offline operation is a key requirement.	Mar 2027
3	**Automated sorting.** Servo/stepper mechanism routes items to the right compartment.	At least 90% correct routing over 20 trials per class (80 trials total).	Early parallel prototyping lowers mechanical risk.	Apr 2027
4	**Remote IoT monitoring.** Flask dashboard shows fill level and status per compartment.	Updates within 5 s on any browser on the local network; alert shown when a bin is nearly full.	Flask and ultrasonic sensors are simple and well documented.	Apr 2027
5	**Performance evaluation.** Benchmark against comparable Raspberry Pi systems.	Accuracy, latency and power reported side by side with at least 3 published systems.	Literature already reviewed in the proposal.	May 2027

7. Stakeholder Analysis

Stakeholders are classified by interest and influence so that communication effort is spent where it matters most.

Stakeholder	Interest	Influence	Approach	Engagement strategy
Project Supervisor (Modassir Ishfaq)	Technical quality, timeline, academic rigour	High	Manage closely	Fortnightly Friday progress meetings; milestone sign-off
Project Team (Sarmad, Rafay, Zeeshan)	Delivery, learning, grades	High	Manage closely	Shared task tracking; defined role ownership; weekly internal syncs

Stakeholder	Interest	Influence	Approach	Engagement strategy
FYP Evaluation Committee	Compliance, novelty, feasibility	High	Manage closely	Formal proposal, progress and final defense presentations
COMSATS Dept. of Computer Engineering	Academic and SDG-alignment standards	Medium	Keep satisfied	Reports submitted per department schedule
End users (simulated in testing)	Usability, sorting reliability, low maintenance	Medium	Keep informed	User-style testing in evaluation phase; feedback used in refinement
Municipal bodies (e.g., LWMC) - indirect	Possible future adopters	Low	Monitor	Contextual motivation only; no engagement in FYP scope

8. Expertise and Skills Plan

The SWOT analysis notes limited prior experience with embedded ML. The table lists the expertise the project needs and how each gap will be closed, so that the skills risk is managed rather than assumed away.

Expertise required	Lead area	How it will be strengthened
Deep learning, dataset preparation, TFLite conversion	Model development (Sarmad)	Public tutorials and papers; supervisor review of each training cycle
Sensors, motors, mechanical design, power	Hardware integration (Rafay)	Early prototyping; datasheets; bench testing after each addition
Raspberry Pi deployment, Flask, system testing	Edge and IoT (Zeeshan)	Incremental builds; documented test cases; peer code review
Project management and technical writing	Whole team	Strategy and Gantt tracking; shared document templates; cross-review of reports

9. SWOT Analysis

Strengths	Weaknesses
• Low-cost single-board hardware keeps the solution affordable • On-device inference removes dependence on unstable internet • Strong prior work (MobileNetV2, TrashNet, TACO, IW-Sorting) lowers technical risk • Alignment with SDG 11, 12 and 13	• Limited team experience with embedded ML and hardware integration • Restricted compute on Raspberry Pi 4 caps model size and accuracy • Mechanical sorting adds reliability risk • Nine-month timeline shared with coursework
Opportunities	**Threats**
• Growing demand for affordable waste solutions in Pakistan • Public datasets reduce data-collection effort • Potential to grow into a deployable product • SDG narrative supports publication or competition entry	• Real-world images differ from datasets (lighting, occlusion, mixed items) • Delays in motors, sensors or camera delivery • Model may miss real-time latency targets • Coursework conflicts across the same semester

Strategic implications. Use the strength of public datasets and pre-trained models to offset limited experience (S-W); start mechanical prototyping early and quantise the model from day one to guard against hardware and

latency threats (W-T); and treat the SDG narrative as the route to a publication or competition entry once the prototype is proven (S-O).

10. Project Scope Strategy

In scope	Out of scope
• On-device classification into four categories: biodegradable, recyclable, non-recyclable, hazardous • Dataset assembly from TrashNet, TACO and locally collected images • Model training, quantisation and TFLite deployment on Raspberry Pi 4 • Servo/stepper mechanical sorting • Ultrasonic fill-level sensing per compartment • Local Flask dashboard with alerts over Wi-Fi • Software and system testing and an evaluation report	• Cloud-based or internet-dependent classification • City-scale or multi-bin fleet deployment • Live integration with municipal systems (e.g., LWMC) • Commercial-grade enclosure, certification or mass production • Mobile application (the interface is browser-based only)

11. Project Strategy, Phases and Deliverables

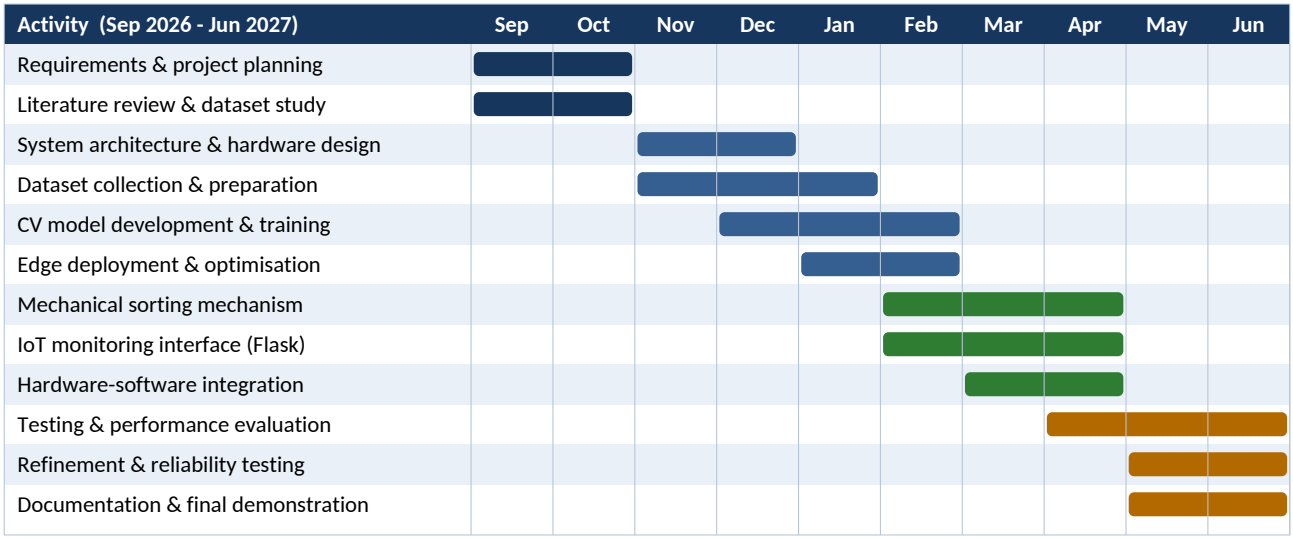
11.1 Execution strategy

The project uses a **hybrid** strategy: a predictive (waterfall-style) structure at the macro level, because FYP milestones and supervisor checkpoints are fixed by the department calendar, and an iterative (agile-style) approach inside each technical phase, where short build-test-refine cycles are needed for model development and integration.

11.2 Phases, milestones and exit criteria

Phase	Period	Approach	Key activities	Milestone / exit criterion
1. Foundation	Sep - Oct 2026	Predictive	Requirements, planning, literature review, dataset study	Scope, requirements and dataset plan approved
2. Build	Nov 2026 - Feb 2027	Iterative	Architecture and hardware design, dataset collection, model training, edge deployment and optimisation	Model meets accuracy and latency targets on Raspberry Pi 4
3. Integrate	Feb - Apr 2027	Iterative	Sorting mechanism, IoT dashboard, hardware-software integration	End-to-end sorting works for all four classes
4. Evaluate and Deliver	Apr - Jun 2027	Predictive	Testing, benchmarking, refinement, documentation, final demonstration	Evaluation report and final defense delivered

11.3 Schedule (Gantt chart)



Friday supervisor meetings (fortnightly) are the main control point for tracking schedule adherence and triggering corrective action when a phase falls behind.

11.4 Key deliverables

Deliverable	Owner	Target
Approved scope, requirements and dataset plan	All members	Oct 2026
Prepared dataset (TrashNet, TACO and local images)	Sarmad	Dec 2026
Quantised model running on Raspberry Pi 4	Sarmad, Zeeshan	Feb 2027
Mechanical sorting prototype	Rafay	Mar 2027
Flask monitoring dashboard	Zeeshan	Apr 2027
Integrated prototype and test results	All members	Apr 2027
Performance evaluation report	All members	May 2027
Final report and demonstration	All members	Jun 2027

Proposal, progress and defense submissions follow the department calendar.

12. Resource Strategy and Budget

12.1 Team roles

Team member	Primary responsibility
Muhammad Sarmad (FA23-BCE-069)	Model development and training: dataset preparation, CNN training, TFLite conversion
Abdur Rafay (FA23-BCE-008)	Hardware and mechanical integration: camera, sensors, servo/stepper sorting mechanism
Zeeshan Ali (FA23-BCE-128)	Edge deployment, Flask IoT dashboard and system testing

Roles are indicative and collaborative; all members contribute to integration and testing.

12.2 Indicative budget

Item	Estimated cost (PKR)
Raspberry Pi 4 (board)	28,000
Pi Camera module	5,000
ToF / ultrasonic sensors	4,000
Servo motor (inlet flap)	1,500
Stepper motor and driver (bin rotation)	4,000
5V/3A regulated power supply	3,000
microSD card, cables, connectors	3,000
Bin body, compartments and mechanical parts	10,000
Wiring, breadboard, miscellaneous	3,000
Test samples, printing and documentation	4,000
Subtotal	**65,500**
Contingency (10%)	6,550
Total (indicative)	**72,050**

Costs are rough estimates for planning and must be checked against current local vendor prices. If the cost is shared equally, it is about PKR 24,017 per member. Software is free and open source (TensorFlow Lite, Flask) and datasets are public.

13. Risk Management Strategy

Likelihood (L) and impact (I) are rated Low = 1, Medium = 2, High = 3; the score is L x I (1-3 low, 4 medium, 6-9 high).

ID	Risk	L	I	Score	Response strategy
R1	Accuracy falls short on real-world images	2	3	6	Mitigate: add locally collected images; augmentation (rotation, brightness, occlusion)
R2	Inference latency too high on Raspberry Pi 4	2	3	6	Mitigate: lightweight model and TFLite quantisation from the start
R3	Sorting mechanism jams or misaligns	2	2	4	Mitigate: early parallel prototyping; low-confidence fallback to unsorted compartment
R4	Hardware component delays	1	2	2	Avoid: order all components early in Phase 1
R5	Schedule conflicts with coursework	2	2	4	Accept/mitigate: fortnightly supervisor cadence to catch slippage early
R6	Scope creep (cloud, mobile app)	1	2	2	Avoid: enforce the scope boundaries in Section 10
R7	Integration issues between modules	2	3	6	Mitigate: incremental integration with testing after each subsystem
R8	Unsafe handling of hazardous test items	1	3	3	Avoid: use harmless look-alike samples or images for the hazardous class

14. Communication Strategy

Audience	Method	Frequency	Purpose
Project Supervisor	In-person or online meeting	Fortnightly (Fridays)	Progress review, guidance, risk escalation
Project Team	Internal sync (in person or chat)	Weekly	Task coordination and resolving blockers
FYP Evaluation Committee	Formal presentation and report	Per milestone	Evaluation and sign-off
Department	Written report submission	Per department deadlines	Compliance and record keeping

15. Quality Management Strategy

Quality is managed against the proposal's three criteria - classification accuracy, inference latency and power consumption - each benchmarked against comparable Raspberry Pi systems in recent literature.

Test level	What is checked	Pass criterion
Software	Accuracy, precision and recall per class on a held-out test set	At least 90% accuracy (Objective 1)
System	End-to-end sorting with real samples under varied lighting and presentation	At least 90% correct routing (Objective 3)
IoT	Fill-level reporting and alert delivery on the local dashboard	Updates within 5 s; alert at near-full level
Documentation	Evaluation report with results, trade-offs and future improvements	Reviewed by supervisor before submission

16. Success Criteria

- A working prototype integrating camera, sensors, actuation and Raspberry Pi 4 is demonstrated at the final defense.
- The model classifies the four target categories with accuracy comparable to published benchmarks.
- The system runs entirely offline with no cloud dependency.
- The dashboard reports live fill-level and status data over the local network.
- All FYP milestones (proposal, progress reviews, final report, demonstration) are delivered on schedule.

17. Conclusion

This strategy turns the Smart Waste Segregation proposal into an actionable management plan. By defining the need, concept, vision and mission, SMART objectives, size, stakeholders, expertise, hybrid strategy, phases, deliverables, budget, risks and quality measures, the team is positioned to deliver a working, evaluable prototype within September 2026 - June 2027, while staying true to its goal: showing that affordable, locally deployable AI hardware can improve source-level waste segregation in resource-constrained environments such as Pakistan.

References

- FYP Proposal Report, "AI-Powered Smart Waste Segregation System Using Edge Computing and IoT," COMSATS University Islamabad, Lahore Campus, Department of Computer Engineering, 2026 (Supervisor: Modassir Ishfaq).
- Public mission and vision statements of Tesla, Microsoft and Google, from the "About" pages of their official websites (paraphrased in Section 4.3).